

Supplementary Material

Case implementation and provenance for the six-gate audit

This supplement accompanies “A Benchmark-Integrity Audit Protocol for Conformal Battery State-of-Health Evaluation.” It makes the retrospective case implementation auditable. The integrated G1–G6 synthesis was developed after the diagnostic program; it was not prospectively registered. The component plan files, scripts, and current paper data are preserved at immutable public replication commit [42f1120ed1f5cbc0282a706ebd4bec24a8d19d6e](https://github.com/42f1120ed1f5cbc0282a706ebd4bec24a8d19d6e).

Table S1. Analysis-to-gate implementation map

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A01 / exp001	Pooled-domain, cell-held-out coverage baseline	G1, G2, G3	Main; governed cell-disjoint train/calibration/evaluation partitions	QA numeric schema; recent permitted only as a declared sequential measurement	Ridge, gradient boosting (GB), random forest, and extra trees with standard, weighted, domain-Mondrian, online-adaptive, and persistence-anchor conformal candidates; selected diagnostic is persistence-anchor protocol-Mondrian

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A02 / exp002	Recent-label and resistance ablation	G1	Main; identical governed partitions across four conditions	Full; minus prev_soh ; minus resistance; minus both; persistence candidate disabled when prev_soh is absent	Same candidate families as A01; validation-only selection within each condition
A03 / exp003	Cross-dataset transfer diagnostic	G1, G2, G3	LCAL, TCAL, LNAS, TNAS, LOxf, TOxf	Cross-domain numeric schema excludes dataset-, chemistry-, and protocol-identity columns; recent label remains a declared condition	Same point-predictor and conformal candidate families as A01, including protocol-Mondrian where groups exist
A04 / exp004	Residual-bias stress screen	Ancillary to G3/G4 interpretation	Main test rows from A01	No refit; additive bias is applied to the selected point prediction	Selected A01 persistence-anchor interval; conformal radius is fixed

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A05 / exp005	Host-timing screen	Ancillary only	512 real Main test rows	prev_soh and precomputed calibration-group radius	Persistence-only lookup, persistence-anchor interval lookup, and GB inference; no deployment claim
A06 / exp006	Method-family and dependence-aware comparison	G2, G3	Main plus all six cross-dataset tasks	Governed task-specific schemas from the split manifest	Persistence-anchor protocol-Mondrian, GB quantile regression (QR), GB conformalized quantile regression (CQR), and NGBoost-CQR
A07 / exp007	Cross-family UQ and two-threshold decision audit	G2, G3, G5	Same seven tasks and corrected disjoint rows	Same task schemas as A06	Persistence-anchor, GB-QR, GB-CQR, and NGBoost-CQR; decision cells at SOH 0.70 and 0.80

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A08 / exp008	Reliability and decision-consequence audit	G1, G2, G3, G5	Same seven tasks and corrected disjoint rows	Declared recent-label availability and task schemas	A07 families; cell-block uncertainty, subgroup checks, false acceptance, false rejection, and uncertainty
A09 / exp009	Schema-cleaning and residual-shift audit	G4, G5	Same seven tasks; added harmonized LNAS diagnostic	Original task schema versus a common cross-dataset schema	GB-CQR and supporting A08 families; identical rows within contrasts
A10 / exp010	No-recent-label hard-regime audit	G1, G4, G5	Seven tasks; horizons 0, 5, and 20; six SOH thresholds	No <code>prev_soh</code> ; original, common-cycle, and stage schemas	GB-QR, GB-CQR, and stage-Mondrian CQR

ID	Governed analysis	Gate(s)	Tasks and rows	Feature schema	Model / conformal family
A11 / exp011	Schema-to-decision audit and independent control	G1, G4, G5	Seven horizon-0 tasks; identical within-task rows; 245 contrast/end-point tests	As-used bundle, clean-original, common-cycle no-recent, common-cycle plus recent-label reference, and clean plus a fixed-seed permuted log-cycle control	One GB-CQR procedure fitted under each schema; paired cell-block deltas
A12 / exp012	Localized-residual sensitivity comparator	G6	Same seven corrected tasks and identical standard/localized rows	Clean task features shared by calibration and target rows; no density-ratio estimate	Standard GB-CQR versus Gaussian-kernel localized-residual CQR; sensitivity diagnostic only

Table S2. Locked parameters, selection, plan provenance, rerun, and deviations

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A01	Seed 42; GB: 250 trees, depth 3, learning rate 0.035; RF/ET: 300 trees, minimum leaf 3; q grid 0.90, 0.93, 0.95, 0.97, 0.99, 1.00	Validation coverage first, then minimum validation width; test metrics excluded	experiments/experiment/ 2026-05-20; e2ed4beba44f49f92d136f3c66a545bd1528eb010	POB1-2026-07-18-01 full rerun; scripts/run_real_experiments.py --seed 42	Legend experiment standalone real experiments.py current manuscript values were regenerated in the pinned full rerun and are covered by the paper SSOT/raw recomputa- tion gate
A02	Seed 42; four predeclared drop-feature conditions; A01 model and q grids	A01 validation- only rule, applied separately within each condition	experiments/experiment/ 2026-05-20; e2ed4beba44f49f92d136f3c66a545bd1528eb010	POB2-2026-07-18-01 full rerun	Preselected model family across conditions; therefore the Main contrast is pipeline sensitivity, not a method- matched feature effect

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A03	Seed 42; A01 model/q grids; target windows 0–20% train, 20–35% calibration, 35–55% validation, >55% test	Online candidate preferred only if validation coverage is at least 0.90; otherwise coverage- first/width selection	experiments/exp03_2026-05-20; 2026-05-20; e2ed4beba44f49692d136f3c66a545bd528eb010	POB3-2026-07-18-01 full rerun	Negative plan.md; transfer retained; no broad- transfer success claim
A04	Bias grid −0.05, −0.02, −0.01, −0.005, 0, +0.005, +0.01, +0.02, +0.05	No tuning or radius refit after bias injection	experiments/exp04_2026-05-20; 2026-05-20; e2ed4beba44f49692d136f3c66a545bd528eb010	POB4-2026-07-18-01 full rerun	Positive plan.md; bias screen only; not evidence of a physical degradation mechanism
A05	20 warmups, 300 timed repetitions, 512 real rows	Fixed timed path; no model selection	experiments/exp05_2026-05-20; 2026-05-20; e2ed4beba44f49692d136f3c66a545bd528eb010	POB5-2026-07-18-01 full rerun	Howd; amortized only; explicitly excluded from edge/deployment evidence

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A06	Seed 42; 1,000 cell-block resamples; q grid 0.90–1.00; GB quantiles 0.05/0.95 with 250 trees, depth 3, learning rate 0.035; NGBoost 300 estimators, learning rate 0.03	Validation coverage first, then width; contribution positioning fixed before result inter- pretation	experiments/experiments/cap06/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912e0051ddc23	experiments/cap06/plan.md; corrected 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe1	experiments/cap06/plan.md; repair/results/r diagnostic rather than method superiority
A07	Seed 42; 1,000 block resamples; q/GB/NGBoost settings as A06; denominator floor 5	Validation coverage-first width; inference intervals do not tune q	experiments/experiments/cap07/plan.md; 2026-05-30; 83d81d8c59c893979c57ee55b62912e0051ddc23	A007ufpa_round12repair/plan.md; manifest; rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe1	experiments/cap07/plan.md; made target validation/tes windows pairwise disjoint and earlier overlapping outputs
A08	Seed 42; 1,000 block resamples; denominator floor 5; A07 method settings	Same-row method comparison; failed cells/tasks retained	experiments/experiments/cap08/plan.md; 2026-06-01; ba5080bc38f97f7a450d19651b22db41005f95b	A008unreliability_audit/plan.md; manifest; rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe1	experiments/cap08/plan.md; preserves multiplicity in cell resampling; remain illustrative decision cutoffs

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A09	Seed 42; 1,000 block resamples; denominator floor 5; q grid 0.90–1.00	Identical rows and coverage- aligned comparisons; common- schema diagnostic added for LNAS	experiments/exp009ufpa_round4repair/plan.md; 2026-06-01; ba5080bc38f9717aaca59d19651b22db141005f95b	manifest; rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe1	schema cannot eliminate residual drift; no physical- mechanism attribution
A10	Seed 42; 1,000 block resamples; horizons 0/5/20; thresholds 0.65/0.70/0.75/0.80/0.85/0.90; denominator floor 5; q grid 0.90–1.00	Validation- only q selection; Bonferroni and BH multiplicity	experiments/exp010uhard_regAnalsudit/plan.md; 2026-06-01; ba5080bc38f9717aaca59d19651b22db141005f95b	manifest; rerun 2026-07-18; code commit d9345592979d2c929e611b54056b4176e79dbe1	no-recent containing bookkeeping fields are contami- nated and superseded by A11 contrasts

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A11	Seed 42; control- column seed 1033; 1,000 paired cell-block resamples; family size $5 \times 7 \times 7 = 245$; family-wise alpha 0.05; denominator floor 5; GB settings as A06	Identical task rows; Bonferroni- adjusted normal intervals from block- bootstrap SE; exact paired-cell sign test at nonzero zero-SE boundaries	experiments/external/plan.md; 2026-06-01; ba5080bc38f97170a59d19651b22db141005f95b	A011 original paper manifest; rerun 2026-07-18; code commit d9345592979d2c929e641b54056b4176e79dbe1	Hydrous substance/plan.md; repair earlier duplicated-fit control with idently permuted log-cycle control. Because that single-draw control is non-null in LCAL/TCAL, it is a veto only and blocks mechanism attribution

ID	Hyperparameters / search grid	Calibration and selection rule	Plan identifier, archive date, immutable plan commit	Governed run identifier	Recorded deviation or claim boundary
A12	Seed 42; 1,000 block resamples; denominator floor 5; bandwidth grid 0.25, 0.50, 1, 2, 4, infinity	Validation- only coverage-first width selection; test metrics excluded	experiments/experiments/2026-06-01; ba5080bc38f9717a0a59d19651b22db41005f95b	A12_manifest; rerun 2026-07-18; code commit d9345592979d2c029e611b54056b4176e79dbe1	Effective_cp_comparator/plan implemented is localized residual weighting, density-ratio or guarantee- bearing shift- adaptive conformal method; one point- coverage failure recovers, no lower-bound shortfall recovers

Provenance boundary

The plan-archive commits above precede the governed July 2026 rerun that supplies the current manuscript values. They establish that component analysis rules existed before the corrected rerun; they do not make the integrated six-gate synthesis prospectively registered. The earlier exploratory processed CSV could not be recovered byte-for-byte. The paper therefore reports only the corrected rerun regenerated from preserved raw holdings under Python 3.11.15, NumPy 2.2.6, pandas 2.3.3, and scikit-learn 1.7.2. Supplementary claims must be interpreted with the same case-bounded scope as the manuscript.